

PHANENDRA PALADUGU

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PROFESSIONAL SUMMARY

Data Scientist with 5+ years of hands-on experience designing, building, and deploying machine learning models, deep learning pipelines, and statistical models using Python, TensorFlow, Scikit-learn, and PySpark across enterprise and financial environments. Strong expertise in exploratory data analysis, feature engineering, predictive modeling, NLP, and computer vision. Experienced in building end-to-end ML workflows from data preprocessing and feature selection through model training, hyperparameter tuning, evaluation (AUC-ROC, precision/recall, F1), and production deployment with drift monitoring. Proven background working with large-scale transactional and financial datasets, building ETL pipelines, and delivering actionable insights through data visualization (Power BI, Tableau, Streamlit). Skilled in collaborating with business stakeholders, risk teams, and cross-functional partners to translate requirements into reliable, production-grade ML solutions.

EXPERIENCE

Data Scientist | Wells Fargo

Jan 2025 - Present

- Build and deploy predictive models for credit risk assessment and customer churn prediction using XGBoost, LightGBM, and Scikit-learn on large-scale financial datasets, improving early default identification accuracy.
- Perform feature engineering and exploratory data analysis on transactional, behavioral, and demographic data using Pandas, NumPy, and PySpark to identify key risk predictors and high-value customer segments.
- Develop model evaluation frameworks incorporating precision/recall, AUC-ROC, lift charts, and confusion matrices; track all experiments using MLflow for reproducibility and version control.
- Build automated data preprocessing and feature extraction pipelines using PySpark and SQL, handling missing values, outlier detection, encoding, and feature scaling on million-row datasets.
- Create interactive dashboards and model performance reports using Streamlit and Power BI to communicate risk scores, forecast trends, and analytical findings to risk management and leadership teams.
- Collaborate with risk management, compliance, and product teams to translate business requirements into data-driven strategies, presenting model outputs and recommendations to senior stakeholders.
- Implement model monitoring and drift detection workflows using Evidently AI to track prediction stability, data distribution shifts, and trigger automated retraining alerts when performance degrades.
- Perform A/B testing and hypothesis testing on customer engagement strategies using statistical methods (t-tests, chi-square, bootstrap) to validate business decisions with quantitative evidence.

Lab Assistant | Purdue University (Indiana)

Aug 2024 - Dec 2024

- Assisted students in performing electrical measurements including voltage, current, impedance, and frequency response using oscilloscopes, multi-meters, and signal generators across multiple lab scenarios.
- Guided students in designing, wiring, and troubleshooting electronic circuits, applying systematic debugging approaches to identify and resolve hardware issues according to lab specifications.
- Reviewed and evaluated lab notebooks and technical reports, providing detailed feedback on data accuracy, documentation quality, analytical reasoning, and proper visualization of experimental results.

Technology Analyst | Infosys (India)

Jan 2020 - Dec 2021

- Designed and deployed a deep learning classification model using Python (TensorFlow/Keras) to auto-categorize 200+ monthly ServiceNow incidents and recommend resolution articles via NLP-based text similarity, reducing manual triage effort by 30%.
- Performed exploratory data analysis and feature engineering on large-scale enterprise datasets using Pandas, NumPy, and PySpark, applying statistical tests, correlation analysis, and dimensionality reduction to identify key predictors.
- Built predictive models using Scikit-learn and XGBoost for demand anomaly detection across supply chain data, enabling early identification of disruptions and improving forecast accuracy through ensemble methods.
- Created data visualizations and automated reporting dashboards using Matplotlib, Seaborn, and Power BI to communicate model performance metrics (AUC-ROC, precision/recall, F1-score) and analytical findings to cross-functional stakeholders.
- Developed a real-time email monitoring application using Python to auto-parse incoming requests, classify query types using rule-based text matching, and generate reference tickets in ServiceNow, enabling 24/7 automated intake processing.
- Partnered with planning teams and business stakeholders to validate data consistency, support KPI tracking, and troubleshoot data anomalies across global operations.

Senior Systems Engineer | Infosys (India)

Jan 2017 - Dec 2019

- Automated data integration and ETL workflows using Python, Selenium, and SAP BW/APO, processing large-scale master and transactional datasets across global supply chain systems and saving 800+ hours of manual effort annually.
- Built structured data pipelines using SAP BW InfoCubes, DataSources, and process chains, ensuring clean, consistent, and version-controlled data feeds for downstream statistical modeling and forecasting.
- Managed large volumes of master and transactional data across SAP APO and MM modules using CIF, performing data quality audits and validation checks to ensure accuracy and synchronization across integrated enterprise systems.

- Wrote complex SQL queries for data extraction, transformation, and validation across SAP BW systems, supporting ad-hoc reporting and business review cycles for global supply chain operations.
- Supported demand and supply planning systems used globally, resolving high-impact data issues through ServiceNow ticketing, monitoring batch jobs, and collaborating with cross-functional stakeholders to maintain system uptime.
- Developed automated Python scripts for data reconciliation and anomaly detection to proactively identify data inconsistencies between source systems and reporting layers before downstream consumption.
- Built structured data flows using SAP BW DataSources and process chains, enabling clean and consistent data input for weekly forecasting reports, KPI dashboards, and downstream analytics.
- Partnered with planning teams to validate data consistency, troubleshoot data anomalies, and support forecasting accuracy across global demand and supply planning operations.

PROJECTS

IntelliDesk: AI-Powered IT Service Desk

Technologies: Python, FastAPI, XGBoost, TF-IDF, Sentence-Transformers, FAISS, MLflow, Streamlit, Evidently AI, Docker, GitHub Actions

- Built an ML ticket-triage system with dual XGBoost classifiers for category and priority; diagnosed and fixed a 12:1 class imbalance where standard balanced weighting hurt F1, using sqrt-scaled weights, alongside error analysis and a sentence-transformer + FAISS retriever, all tracked in MLflow with Evidently drift monitoring and a 39-check validation suite. [\[Link\]](#)

DocBrain: Enterprise Document Intelligence Engine

Technologies: Python, PyTorch (CUDA), ChromaDB, Sentence-Transformers, Llama 3.1 8B, Tesseract/DocTR, SpaCy, FastAPI, Streamlit, Docker

- Built a grounded RAG pipeline over 4 document types with ChromaDB, Sentence-Transformers, and Llama 3.1 8B; ran a quantitative eval on 30 hand-written queries (0.90 Recall@5, 0.78 MRR, 1.00 Recall@1 on contracts) plus a second-layer LLM-judge faithfulness score that surfaced answer-quality issues on correctly-retrieved queries, with quality-based OCR routing and SpaCy NER. [\[Link\]](#)

EDUCATION

Master of Science - Electrical and Computer Engineering GPA: 3.61/4
Purdue University Northwest, Hammond, Indiana 2025

Coursework: Machine Learning, Neural Networks, Big Data, Computer Vision, Digital Signal Processing, Computer Networks

TECHNICAL SKILLS

Languages	Python, SQL, R, MATLAB, Bash, ABAP
ML / DL Frameworks	TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, LightGBM, PySpark, Hugging Face Transformers
NLP & Computer Vision	NLTK, SpaCy, Sentence Transformers, OpenCV, FAISS, ChromaDB, Transfer Learning, OCR (Tesseract/DocTR)
Data Engineering	Pandas, NumPy, Apache Spark, Hadoop, ETL/ELT Pipelines, MySQL, Teradata, SAP BW/APO, Data Modeling
Statistical Methods	Regression, Classification, Clustering, Time Series Forecasting, A/B Testing, Hypothesis Testing, EDA, Dimensionality Reduction, Bayesian Methods
Visualization & BI	Matplotlib, Seaborn, Plotly, Streamlit, Power BI, Tableau, Excel (Pivot Tables, VLOOKUP, Charting)
MLOps & Deployment	FastAPI, Flask, Docker, Git, CI/CD, MLflow, Evidently AI, pytest, GitHub Actions, LangChain, Model Registry
Cloud & Platforms	AWS (S3, SageMaker, EC2), GCP (BigQuery, Vertex AI), Llama 3.1, ServiceNow
Collaboration & Tools	Agile/Scrum, JIRA, Confluence, Cross-Functional Partnership, Stakeholder Communication, Technical Documentation

KEY ACHIEVEMENTS

- Reduced IT ticket resolution time by 30% through automated NLP-based classification and knowledge retrieval system, processing 200+ incidents per month across global enterprise support operations.
- Saved 800+ hours annually by automating data integration, ETL workflows, and reporting pipelines using Python, Selenium, and SAP BW/APO across global supply chain systems.
- Built 2 production-grade open-source ML systems with full CI/CD pipelines, comprehensive test suites, and drift monitoring: IntelliDesk (39 validation checks) and DocBrain (162 pytest tests, 0.90 Recall@5, 1.00 Recall@1 on contracts).
- Achieved 96.27% mean IoU on satellite road segmentation using U-Net (TensorFlow/OpenCV) and improved rare network attack detection by 18% through transfer learning on the UNSW-NB15 dataset (250K+ samples).